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Write a program for Implementation of K-Nearest Neighbours (K-NN) in Python

CODE:

```
import math

train = [[2,4,'A'], [4,6,'A'], [6,8,'B'], [8,10,'B']]
test = [5,7]
k = 3
distance = []
for i in train:
    d = math.sqrt((i[0]-test[0])**2 + (i[1]-test[1])**2)
    distance.append([d, i[2]])
distance.sort()
a = 0
b = 0
for i in range(k):
    if distance[i][1] == 'A':
        a += 1
    else:
        b += 1
if a > b:
    print("Predicted Class: A")
else:
    print("Predicted Class: B")
```

OUTPUT

main.py	Output
<pre>1 import math 2 train = [[2,4,'A'], [4,6,'A'], [6,8,'B'], [8,10,'B']] 3 test = [5,7] 4 k = 3 5 distance = [] 6 for i in train: 7 d = math.sqrt((i[0]-test[0])**2 + (i[1]-test[1])**2) 8 distance.append((d, i[2])) 9 distance.sort() 10 a = 0 11 b = 0 12 for i in range(k): 13 if distance[i][1] == 'A': 14 a += 1 15 else: 16 b += 1 17 if a > b: 18 print("Predicted Class: A") 19 else: 20 print("Predicted Class: B")</pre>	<pre>Predicted Class: A === Code Execution Successful ===</pre>